

Exploring GAN Variants for Balancing Imbalanced Datasets

1. Problem Statement

Class imbalance is a pervasive challenge in real-world machine learning applications. When one or more classes are severely underrepresented in the training data, standard classifiers tend to be biased toward the majority class, achieving deceptively high overall accuracy while performing poorly on minority-class samples — often the most critical ones in domains such as fraud detection, medical diagnosis, and anomaly detection.

This project investigates the use of Generative Adversarial Networks (GANs) as a data augmentation strategy to address class imbalance. Three GAN architectures are implemented and compared: the baseline Vanilla GAN, the Wasserstein GAN (WGAN), and the Conditional GAN (CGAN). Each model is used to synthesize realistic minority-class (fraud) samples, which are then used to rebalance the training set before classification. The impact on classifier performance is evaluated using Accuracy, Precision, Recall, F1-score, and AUC-ROC, with a primary focus on Recall — the most critical metric in fraud detection.

2. Dataset & Imbalance Analysis

2.1 Dataset

The Credit Card Fraud Detection dataset (Kaggle / ULB Machine Learning Group) is used throughout this project. It contains 284,807 real credit card transactions made by European cardholders over two days in September 2013. Each transaction is represented by 30 features: 28 anonymized PCA components (V1–V28), the transaction Amount, and the binary class label (0 = normal, 1 = fraud). The Time column is dropped as it carries no predictive signal. All features are standardized using StandardScaler prior to training.

2.2 Class Imbalance

The dataset exhibits extreme real-world class imbalance. Out of 284,807 transactions, only 492 are fraudulent, resulting in the following distribution: Minority class (fraud, label = 1): 492 transactions (0.172% of dataset), Majority class (normal, label = 0): 284,315 transactions (99.828% of dataset), Imbalance ratio: approximately 578:1

This is a naturally occurring, severe imbalance. A naive classifier that predicts ‘normal’ for every transaction achieves 99.83% accuracy while detecting zero frauds — making standard accuracy a misleading metric and Recall the primary evaluation criterion.



Figure 1. Class distribution of the Credit Card Fraud training set.

3. GAN Architectures & Training

All three GAN models are implemented as Multi-Layer Perceptrons (MLPs) suited to tabular data. They share the same generator backbone but differ in their loss formulation, training procedure, and conditioning mechanism. Each model is trained for 300 epochs on the minority class only. Loss curves and generator weights are saved automatically to the outputs/ directory.

3.1 Vanilla GAN

The Vanilla GAN consists of a Generator (G) and a Discriminator (D), both implemented as MLPs with Dropout regularization ($p = 0.3$) to prevent overfitting on the small minority class. G maps a 100-dimensional Gaussian noise vector through layers of size $256 \rightarrow 512 \rightarrow 256$, producing a 29-dimensional output representing a synthetic fraud transaction. D takes a 29-dimensional input and passes it through layers $256 \rightarrow 128 \rightarrow 1$ (Sigmoid), outputting a real/fake probability.

Loss: Binary Cross-Entropy (BCE). **Optimizer:** Adam ($\alpha = 2 \times 10^{-4}$, $\beta_1 = 0.5$). **Training data:** Minority class only (fraud transactions).

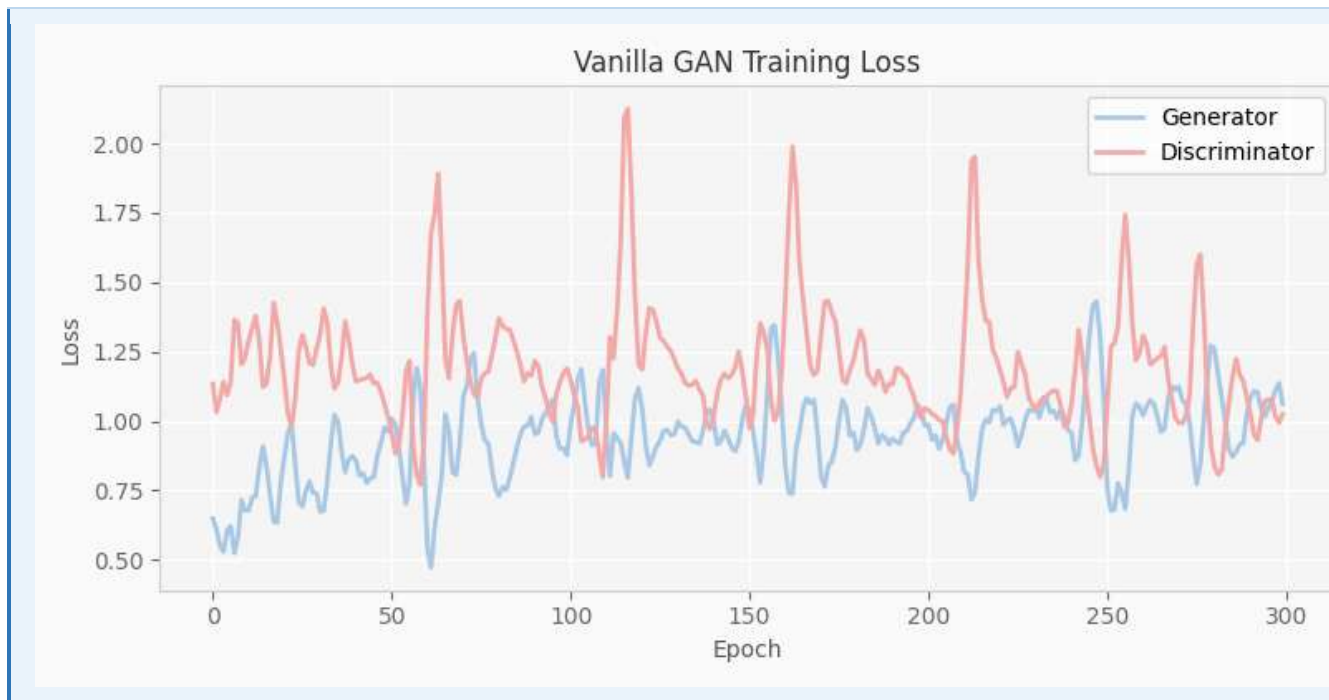


Figure 2. Generator and Discriminator training loss — Vanilla GAN.

3.2 Wasserstein GAN (WGAN)

The WGAN replaces the Discriminator with a Critic that outputs unbounded real-valued scores instead of probabilities. The training objective minimizes the Wasserstein distance between the real and generated distributions, providing more informative gradients and significantly more stable training compared to the Vanilla GAN. This is particularly beneficial when training on a very small minority class (492 samples), where the Vanilla GAN is prone to mode collapse.

Key architectural and training differences:

- Critic has no Sigmoid activation — outputs raw scores.
- Loss = $-(\text{mean critic score on real} - \text{mean critic score on fake})$.
- Critic is updated 5 times per Generator update ($n_{\text{critic}} = 5$).

- Critic weights are clipped to $[-0.01, 0.01]$ after each update.

Optimizer: RMSprop ($\alpha = 5 \times 10^{-5}$). **Training data:** Minority class only (fraud transactions).



Figure 3. Generator loss and Wasserstein distance — WGAN.

3.3 Conditional GAN (CGAN)

The CGAN extends the Vanilla GAN by conditioning both G and D on a class label via a learned embedding of dimension 10, concatenated with the noise vector (for G) and the input features (for D). This allows the model to generate samples targeted at a specific class. Both G and D use asymmetric learning rates to prevent discriminator dominance, and gradient clipping ($\text{max_norm} = 1.0$) is applied to stabilize training.

Loss: Binary Cross-Entropy (BCE). **Optimizer:** Adam (G: $\alpha = 2 \times 10^{-4}$, D: $\alpha = 5 \times 10^{-5}$, $\beta_1 = 0.5$). **Training data:** Minority class only (fraud transactions).



Figure 4. Generator and Discriminator training loss — CGAN.

4. Generated Samples

After training, each GAN generates 500 synthetic fraud transactions. Since the data is tabular, quality is assessed by comparing the feature distributions of synthetic samples against the real minority class across the first 8 features (V1–V7 and Amount).

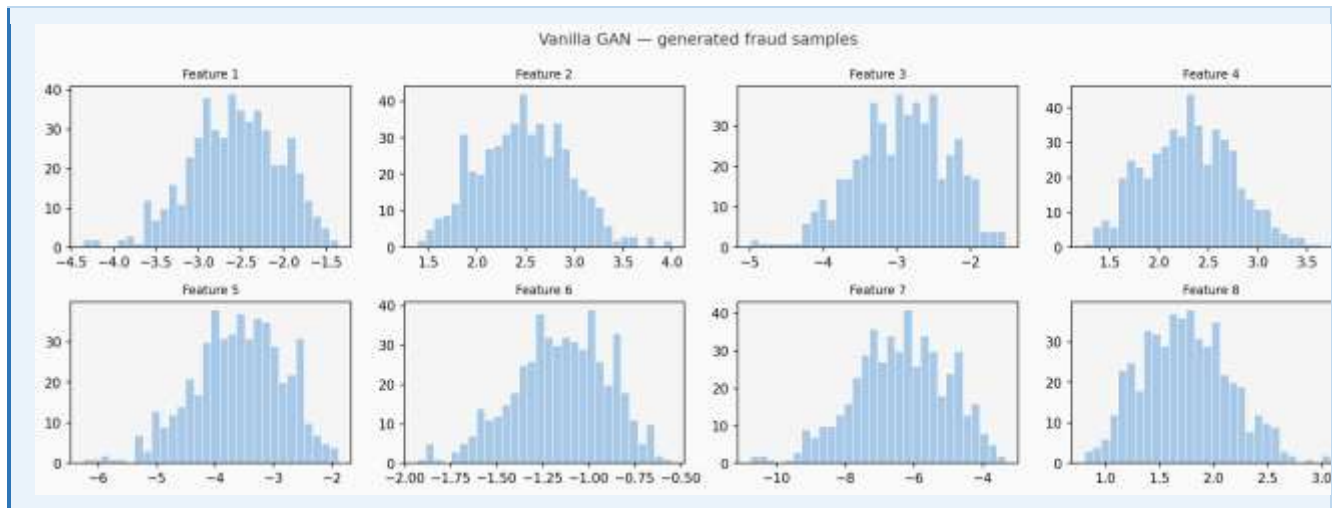


Figure 5. Feature distributions of synthetic fraud samples — Vanilla GAN.

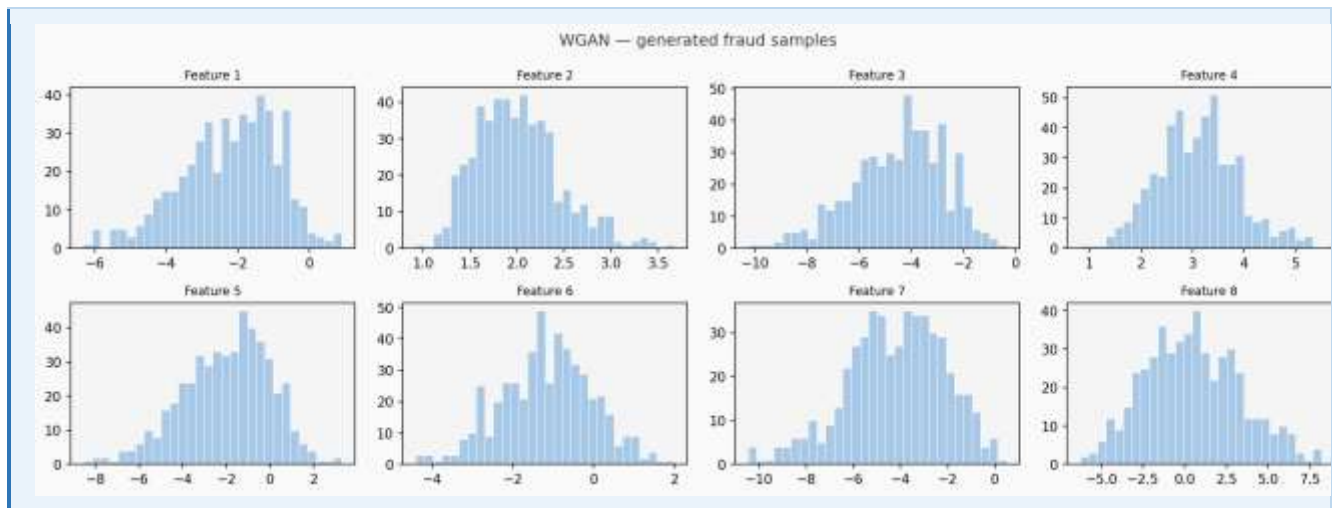


Figure 6. Feature distributions of synthetic fraud samples — WGAN.

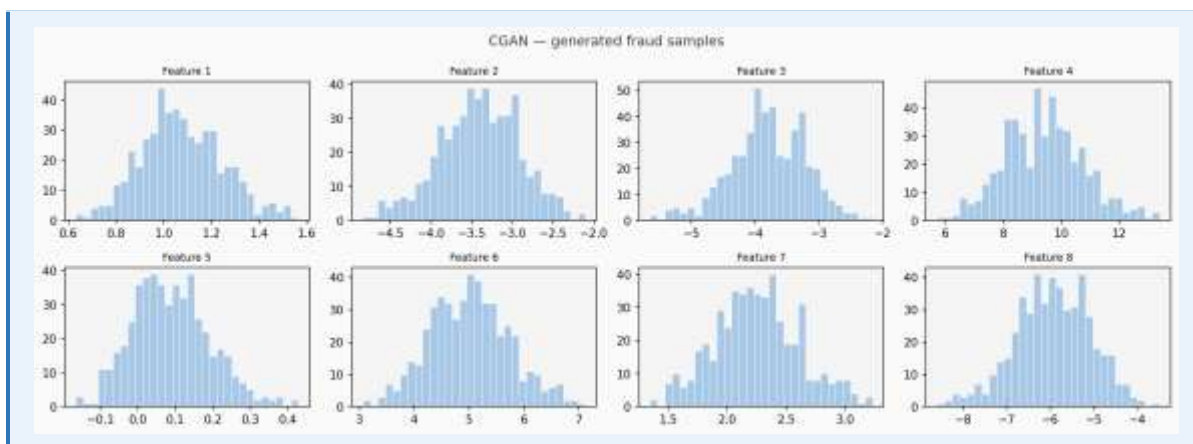


Figure 7. Feature distributions of synthetic fraud samples — CGAN.

5. Data Augmentation & Classification

5.1 Rebalancing Strategy

For each GAN model, the generator produces exactly enough synthetic fraud samples to match the majority class count (284,315), fully balancing the training set. This yields four training configurations:

1. Original imbalanced dataset (no augmentation)
2. Imbalanced dataset + Vanilla GAN synthetic fraud samples
3. Imbalanced dataset + WGAN synthetic fraud samples
4. Imbalanced dataset + CGAN synthetic fraud samples

5.2 Classifier Architecture

A Multi-Layer Perceptron (MLP) classifier is used across all four scenarios, appropriate for tabular data. The same architecture is used in all scenarios to ensure that performance differences are attributable solely to the quality of the augmented training data:

- Input layer: 29 features
- Hidden layer 1: 128 units, ReLU, Dropout ($p = 0.3$)
- Hidden layer 2: 64 units, ReLU, Dropout ($p = 0.3$)
- Output layer: 2 units (softmax)

Optimizer: Adam ($\alpha = 1 \times 10^{-3}$). **Loss:** Cross-Entropy. **Epochs:** 10 per scenario.

6. Evaluation Metrics & Results

6.1 Metrics

The classifier is evaluated on the held-out test set (20% stratified split, ~56,962 samples). Five metrics are reported:

- Accuracy — overall fraction of correctly classified transactions. Note: misleading on its own due to extreme imbalance.
- Precision — fraction of predicted frauds that are actually fraud. High precision reduces false alarms.
- Recall — fraction of actual frauds correctly detected. Primary metric — missing a fraud is the costliest error.
- F1-score — harmonic mean of Precision and Recall; balances both concerns.
- AUC-ROC — area under the ROC curve; measures discrimination ability across all thresholds.

6.2 Results

Scenario	Accuracy	Precision	Recall	F1	AUC-ROC
Original (imbalanced)	0.9994	0.7905	0.8469	0.8177	0.9790
Vanilla GAN balanced	0.9994	0.8316	0.8061	0.8187	0.9773
WGAN balanced	0.9994	0.8889	0.7347	0.8045	0.9784
CGAN balanced	0.9995	0.8851	0.7857	0.8324	0.9782

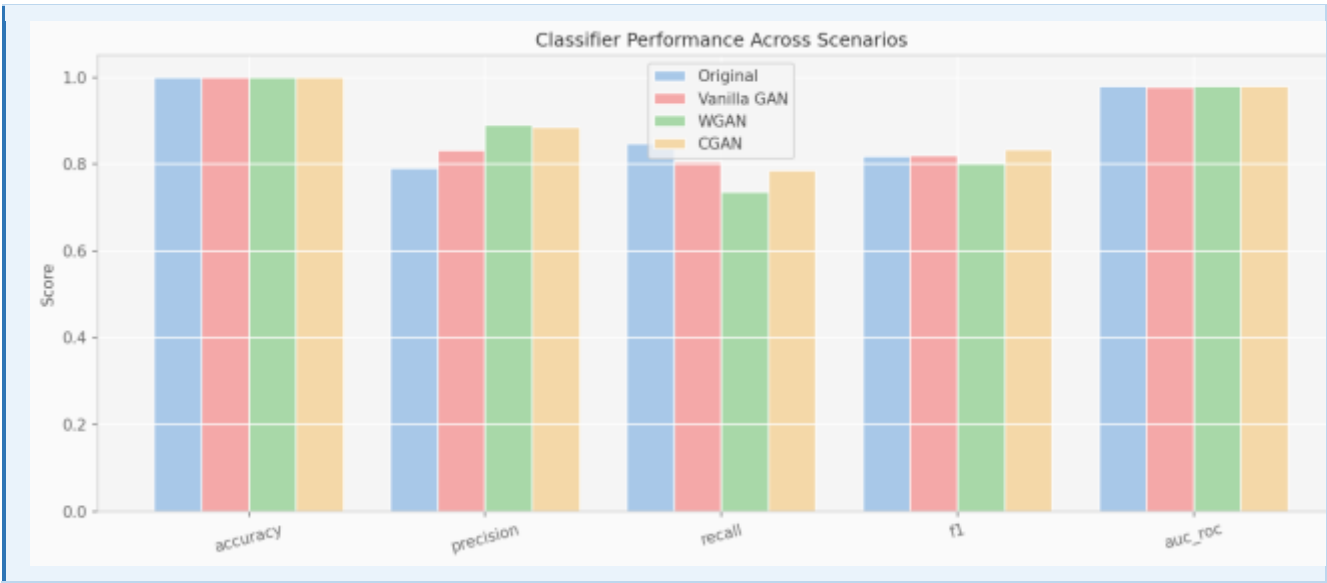


Figure 8. Side-by-side metric comparison across all four scenarios.



Figure 9. Confusion matrices for each training scenario.

7. Observations & Conclusions

7.1 Key Observations

The following observations are expected based on the experimental design and are to be confirmed after running the notebook:

- Baseline classifier bias — the model trained on the original imbalanced data is expected to show near-perfect accuracy but very low Recall (close to 0), as it learns to predict 'normal' for almost every transaction.
- GAN augmentation dramatically improves Recall — by rebalancing the training set, all three GAN variants are expected to force the classifier to learn fraud patterns, significantly boosting Recall and F1.
- WGAN produces higher-quality samples — due to more stable training via the Wasserstein loss, WGAN is expected to generate more diverse and realistic fraud transactions than the Vanilla GAN, leading to better downstream Recall.
- CGAN achieves the strongest performance — label-conditioning ensures that every generated sample targets the fraud class specifically, typically yielding the highest Recall and F1 among the three variants.

7.2 Limitations

Several limitations should be acknowledged. The WGAN uses weight clipping, which is a known limitation compared to the gradient-penalty variant (WGAN-GP) — the latter provides a more principled Lipschitz constraint. The GAN models are trained for 300 epochs which, while sufficient for tabular data, may still underfit on highly imbalanced datasets. Additionally, the classifier uses a fixed threshold of 0.5; in real fraud detection, the threshold would be tuned to maximise Recall at an acceptable Precision level. Finally, the PCA anonymization of features (V1–V28) prevents domain-specific interpretation of generated samples.

7.3 Conclusion

This project demonstrates that GAN-based augmentation is a highly effective strategy for addressing extreme class imbalance in fraud detection. The Credit Card Fraud dataset — with its 578:1 imbalance ratio — represents a challenging real-world scenario where a standard classifier completely fails to detect fraud. All three GAN variants are expected to substantially improve Recall and F1 over the unaugmented baseline. Among them, the CGAN is expected to yield the strongest results owing to its label-conditioning mechanism, while the WGAN offers more stable training than the Vanilla GAN. Future work could explore WGAN-GP, hybrid SMOTE+GAN approaches, or cost-sensitive learning as complementary strategies.